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Capstone Project Synopsis

On

Driving School Management System

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Bachelor of Computer Applications

By:

Radheya Bharat Khandelwal : Div : B with Roll no. B 27

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Mr. Sayyed Saad B
Project Guide

Dr. V.V. Bhosale
HOD

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Abstract

The Driving School Management System is a comprehensive web-based application developed to automate, simplify, and streamline the administrative and operational workflows of modern motor driving schools. Traditional driving academies rely heavily on paper registers, manual logbooks, phone calls, and handwritten fee slips to coordinate student admissions, instructor allocations, driving practice slot schedules, and RTO licensing examinations. These manual practices lead to scheduling conflicts, double-booked training vehicles, misplaced learner records, and lack of real-time progress visibility for learners.

The proposed software system eliminates these operational challenges by offering an integrated online platform. It enables prospective learners to browse available driving packages (such as Two-Wheeler, Four-Wheeler LMV, and Commercial driving packages), register their credentials and Learner License (LLR) details, book convenient practical driving slots, monitor daily attendance, and track syllabus milestone completion. Concurrently, driving school administrators can manage instructor rosters, maintain vehicle fleet maintenance logs, review simulated fee collections, evaluate mock driving test scores, and schedule official RTO driving license tests. The system is engineered using React.js for the responsive frontend, Node.js and Express.js for RESTful API logic, and MySQL for relational data management.

Introduction

In recent years, the rapid growth of automotive transportation and stringent government traffic regulations have heightened the demand for certified, professional driver training institutes. Learning to operate a motor vehicle safely requires a structured blend of theoretical traffic education, simulator familiarization, and supervised on-road behind-the-wheel practice. However, most motor training schools in regional centers still function through fragmented manual record-keeping systems. Physical student appointment cards, paper attendance diaries, and unorganized instructor shifts frequently cause scheduling disputes, underutilized training cars, and lost documentation.

The Driving School Management System is engineered to address these challenges by delivering an end-to-end digital management solution. By establishing a role-based, multi-user web portal for students, driving instructors, and administrators, the system automates student registration, course package selection, practical driving session reservations, and simulated fee collection workflows. Built on modern web standards utilizing React.js, Node.js, Express.js, and MySQL, the application ensures high data integrity, streamlined operational schedules, transparent student evaluation, and seamless preparation for Regional Transport Office (RTO) driving license assessments.

Project Module

Module-I: Student Registration & Authentication Module

Handles secure user registration and login for both learners and administrative staff. Implements password hashing via bcryptjs, session token generation using JSON Web Tokens (JWT), and profile maintenance for capturing personal details, contact numbers, and Learner License (LLR) credentials.

Module-II: Driving Courses & Vehicle Category Catalog Module

Allows learners to explore diverse driver training programs including Two-Wheeler (MCWG), Four-Wheeler (LMV - Manual & Automatic), and Commercial Vehicle driving courses. Displays complete syllabus breakdowns, mandatory theory hours, practical road training hours, and fee structures.

Module-III: Instructor & Vehicle Fleet Scheduling Module

Maintains detailed records of certified driving instructors (licenses, contact info, working hours) and training vehicles (registration numbers, vehicle models, dual-control equipment, fitness and insurance dates). Prevents overbooking and ensures balanced instructor allocation.

Module-IV: Practical Training Session Booking & Attendance Module

Provides an intuitive calendar interface where enrolled students can reserve convenient daily practical driving time slots with assigned instructors. Instructors and admins log real-time daily session completion and biometric/digital attendance records.

Module-V: Evaluation, Mock Test & RTO Test Scheduling Module

Allows instructors to evaluate learner driving competencies across key skills (parallel parking, clutch control, gear shifting, traffic signals, reversing). Facilitates automated online traffic rules mock quizzes and records official RTO driving license test dates and results.

Module-VI: Fee Receipt & Administrative Report Exporting Module

Processes course fee payments through simulated transaction loops, generates instant digital invoice receipts, and enables administrators to generate and export student enrollment lists, instructor performance summaries, and financial reports into downloadable PDF and Excel formats.

Project Plan

Gantt chart:

Sr. No	Task Name	6-July	20-July	Aug-20	Sep-5	Oct-5
1	Requirement Gathering					
2	Planning					
3	Designing					
4	Coding/Development					
5	Testing and Deployment					

Project Requirements:

Hardware Requirements:

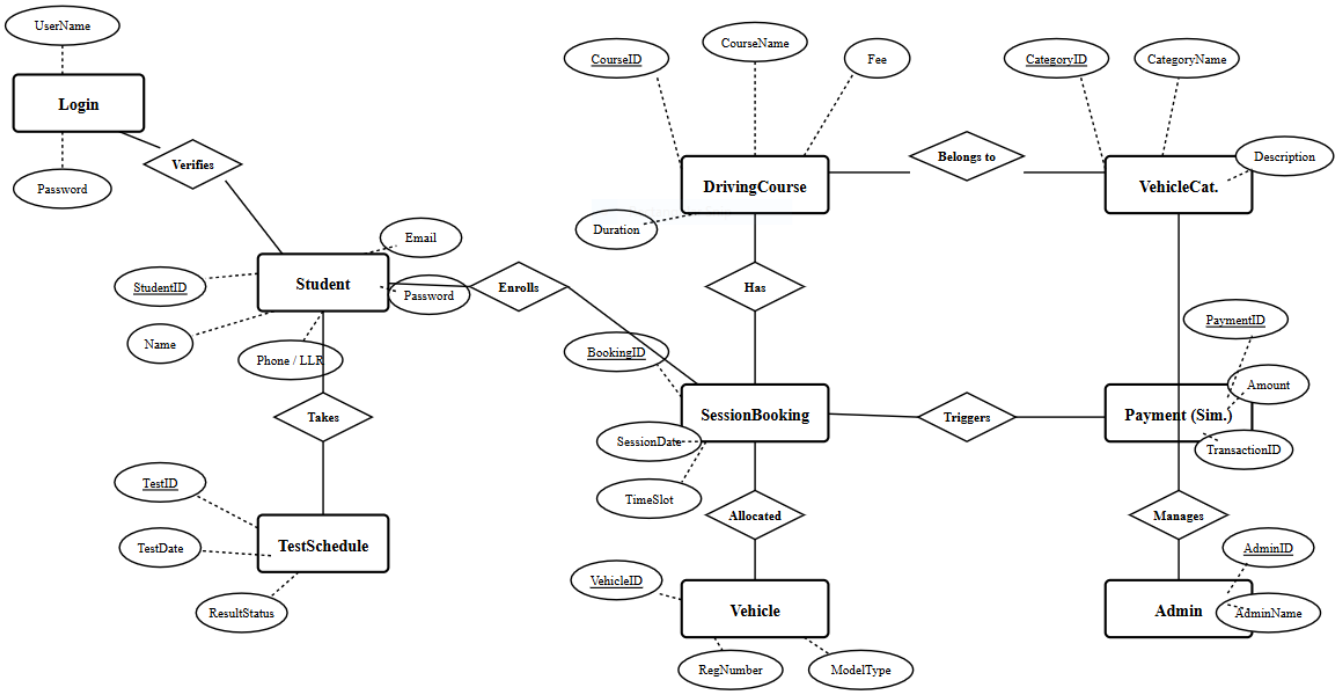
- Processor: Intel Core i3 and above
- RAM: 4 GB and above
- Hard Disk: 160 GB or above
- Input Device: Keyboard, Mouse
- Output Device: Monitor or LCD/LED,
- Printer Network: LAN Network or Internet Connection

Software Requirements:

- Frontend-IDE: HTML5, CSS3, JavaScript, Bootstrap or Tailwind CSS, React.js, VS Code
- Programming Language: Node.js, Express.js
- Backend-Database: MySQL
- Platform: Web Based Application

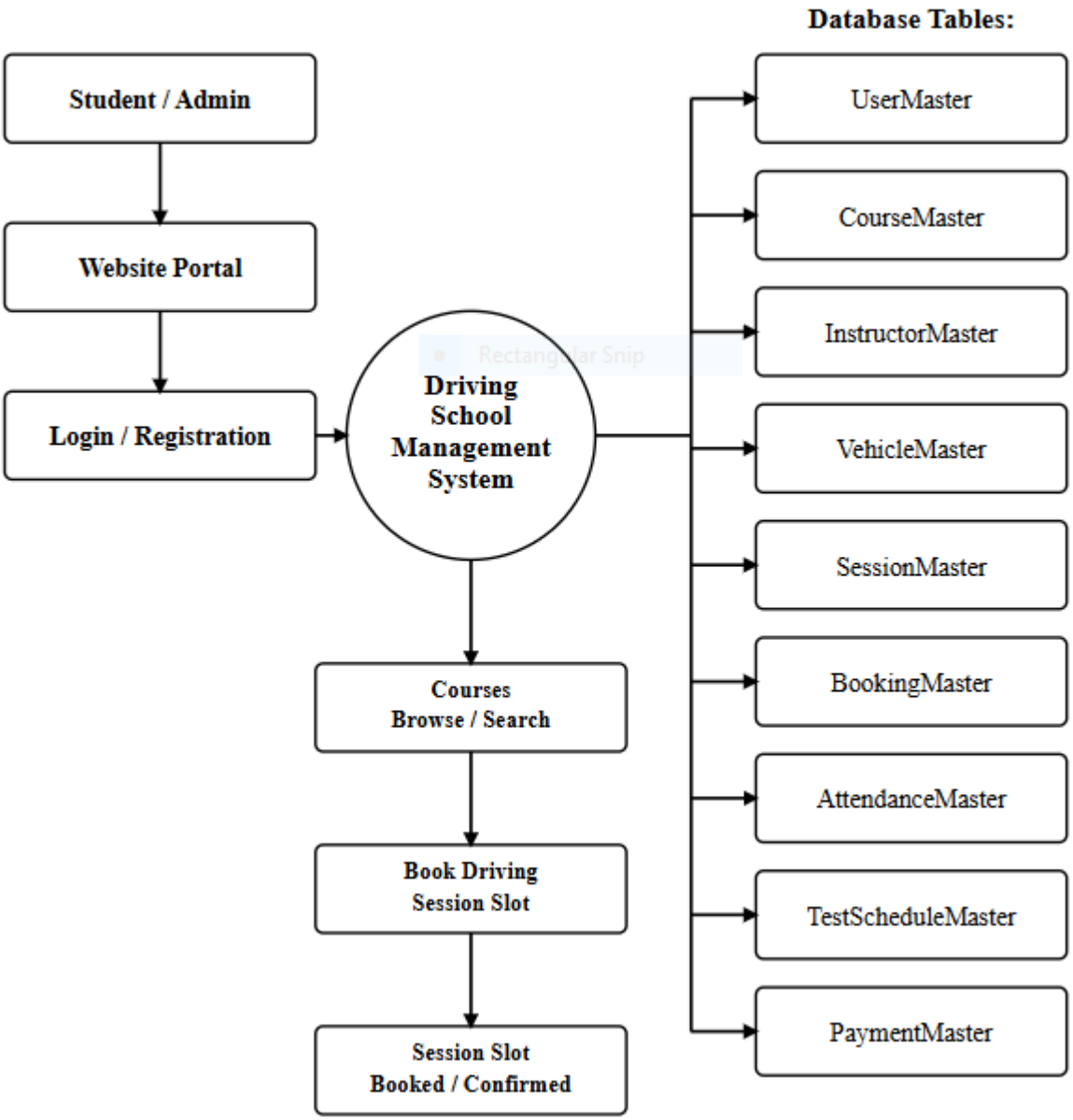
E R Diagram

The Entity-Relationship Diagram below illustrates the relational structure between Students, Courses, Instructors, Vehicles, Practical Sessions, Bookings, Test Schedules, and Payments within the system:



Data Flow Diagram

The DFD Context Diagram below illustrates data flows between external entities and the system:



Conclusion

The Driving School Management System successfully provides a secure, centralized, and user-friendly web platform designed to streamline driver training education and administrative management. By integrating React.js for the responsive frontend, Node.js and Express.js for REST API management, and MySQL for relational data persistence, the project delivers an efficient system that automates learner enrollments, instructor assignments, driving time-slot scheduling, and simulated fee collections.

The project overcomes the inefficiencies inherent in manual paperwork, lost physical appointment slips, and uncoordinated vehicle fleets. It provides learners with transparent insight into their practical training progression, lesson attendance, and mock test scores, while equipping school administrators with comprehensive operational oversight over vehicle maintenance schedules and instructor workloads. The platform can be further extended by incorporating features such as automated SMS/WhatsApp lesson reminders, GPS telematics route replay for training cars, and direct API integration with government RTO Sarathi portals for driving test scheduling.

In conclusion, the Driving School Management System represents a robust, scalable, and modern software solution tailored to meet the needs of contemporary motor training institutes. It showcases how modern full-stack web technologies can enhance administrative efficiency, improve learner safety and competency, and establish an organized foundation for driver education in compliance with modern transport standards.

Bibliography

Books

1. Pressman, Roger S. *Software Engineering: A Practitioner's Approach*. 8th ed., McGraw-Hill Education, 2014.
2. Tahaghoghi, Saied M.M., and Hugh E. Williams. *Learning MySQL*. 1st ed., O'Reilly Media, 2006.
3. Accomazzo, Anthony, Ari Learner, and Nate Murray. *Fullstack React: The Complete Guide to ReactJS and Friends*. 1st ed., Fullstack.io, 2017.

Websites

1. http://www.srtmun.ac.in
2. http://www.cocsit.org.in
3. https://react.dev
4. https://expressjs.com